

YTTEROYANE FYR, Norway

WMO: 013040

Lat: 61.572N

Lon: 4.682E

Elev: 82

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.3	29.7	12.6	10.5	30.5	15.9	12.3	32.5	48.9	40.5	45.6	39.3	11.6	110	0.834

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	68.3	61.3	65.7	59.8	63.5	58.4	62.6	67.2	60.8	64.5	59.1	62.5	8.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.4	78.6	65.4	58.8	74.2	63.6	57.1	69.8	61.7	28.3	67.7	27.0	64.9	25.8	62.7	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
43.5	38.9	35.2	DB	25.8	73.5	3.6	5.6	23.2	77.5	21.1	80.7	19.0	83.8	16.4	87.9
			WB	22.4	64.2	3.6	2.6	19.8	66.0	17.7	67.5	15.7	69.0	13.1	70.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	39.0	38.1	39.2	43.3	47.8	51.9	57.0	58.9	55.3	49.6	44.7	40.8	(6)
	DBStd	8.21	4.12	4.26	3.52	4.05	4.43	3.58	4.50	3.59	3.61	4.22	4.58	4.97	(7)
	HDD50	1841	341	332	336	206	97	18	1	0	3	57	163	287	(8)
	HDD65	6516	806	752	801	651	534	393	254	194	293	478	608	752	(9)
	CDD50	811	0	0	0	5	28	76	218	275	160	44	5	0	(10)
	CDD65	10	0	0	0	0	0	0	6	4	0	0	0	0	(11)
	CDH74	12	0	0	0	0	0	0	12	0	0	0	0	0	(12)
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14) Wind	WSAvg	16.8	20.3	19.2	17.0	14.4	14.0	13.4	12.8	14.0	17.1	18.5	19.7	20.9	(14)
(15) Precipitation	PrecAvg	76.7	7.6	6.5	6.1	4.6	3.6	4.1	4.7	5.5	7.4	8.8	8.6	9.0	(15)
	PrecMax	97.8	13.6	12.1	12.5	8.3	6.1	8.4	7.5	11.0	14.0	15.7	14.7	16.1	(16)
	PrecMin	51.0	1.1	0.0	0.9	0.9	0.7	0.6	0.8	1.8	2.4	3.9	2.0	2.0	(17)
	PrecStd	12.3	3.9	3.2	2.9	1.7	1.5	2.0	1.7	2.3	3.5	3.3	3.9	4.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	47.1	47.3	59.9	64.8	65.8	74.2	71.2	66.9	60.8	55.5	51.0	(19)
		MCWB	43.4	43.2	41.6	51.6	55.1	56.0	64.0	62.2	58.7	53.0	49.9	47.2	(20)
	2%	DB	46.6	45.5	45.5	53.8	59.4	61.2	69.2	68.0	64.2	57.9	53.0	49.4	(21)
		MCWB	43.7	42.9	41.7	46.4	52.5	54.9	62.6	61.3	57.6	52.0	49.1	46.4	(22)
	5%	DB	45.6	44.7	44.6	50.1	55.9	58.3	65.6	65.8	61.9	56.3	51.7	48.4	(23)
		MCWB	43.1	42.5	41.9	44.8	51.1	54.0	60.7	60.5	56.5	51.5	47.6	45.7	(24)
	10%	DB	44.6	43.7	43.6	48.0	53.3	56.2	62.8	63.8	60.1	54.9	50.4	47.3	(25)
		MCWB	42.2	41.6	41.1	43.7	49.4	53.0	58.9	59.3	55.7	50.8	47.0	44.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.3	44.9	44.6	53.6	57.0	57.8	65.3	64.3	61.0	55.8	52.5	48.9	(27)
		MCDB	46.8	45.4	45.1	57.6	62.5	63.0	71.7	68.6	65.1	58.0	53.5	50.0	(28)
	2%	WB	45.1	44.0	43.6	47.6	53.8	55.8	63.0	62.8	59.0	54.3	50.8	47.6	(29)
		MCDB	45.9	44.8	44.5	51.6	57.8	59.5	68.3	66.9	62.2	56.7	52.3	48.9	(30)
	5%	WB	43.9	43.1	42.6	45.8	51.4	54.4	60.9	61.2	57.5	53.0	49.3	46.4	(31)
		MCDB	45.0	44.4	43.8	49.1	55.0	57.5	65.3	64.7	60.6	55.6	51.0	48.0	(32)
	10%	WB	42.6	41.9	41.6	44.5	49.7	53.2	58.7	59.7	56.3	51.3	47.6	44.9	(33)
		MCDB	44.2	43.5	43.2	46.9	52.8	55.9	61.9	62.9	59.4	54.0	49.9	46.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.1	4.5	5.0	5.2	4.5	4.9	4.7	4.4	4.3	4.6	4.6	(35)
		MCDBR	5.1	4.8	4.8	8.1	9.2	8.0	9.2	7.1	6.5	5.2	5.5	5.5	(36)
	5% WB	MCWBR	5.0	4.5	4.2	4.4	5.3	4.8	5.1	4.2	4.1	4.4	4.9	5.7	(37)
		MCWBR	4.9	4.9	4.3	7.0	8.5	7.0	9.0	6.1	5.7	5.1	5.3	5.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.296	0.331	0.351	0.374	0.363	0.358	0.372	0.366	0.355	0.331	0.296	0.365	(40)
		taud	2.098	2.236	2.312	2.321	2.424	2.473	2.459	2.472	2.472	2.378	2.167	2.376	(41)
	Ebn at Noon		121	198	240	258	272	276	268	260	240	199	121	73	(42)
		Edh at Noon	12	21	27	33	32	31	31	28	24	18	11	8	(43)
(44) All-Sky Solar Radiation	RadAvg		56	204	527	1048	1524	1616	1472	1107	640	307	87	34	(44)
	RadStd		10	39	83	120	112	192	127	126	102	50	18	4	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (36 neighbors)		N/A	+1.22	N/A	N/A	N/A	N/A	-166	-243	N/A	N/A

Nomenclature: See separate page